

Quality, Presence, Empathy, Attitude, and Attention in 360-degree Videos for Immersive Communications

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ABSTRACT

Most literature has already shown that virtual reality, concretely, 360-degrees video can have influence on affective skills in multiple applications from education to medicine. This thesis aims to go one step further, real-time 360-degree video communications. Researchers commonly based their experiments on controlled environments and participants with specific characteristics which reduce the reproducibility and ecological validity of the methodologies. We propose to design a methodology based on questionnaires and the use of biosensors to jointly assess technical and socioemotional features in bidirectional immersive communications. For that, we first present some research questions and then choose the tools, stimuli, and use case to answer them. The use case selected as starting point to validate the methodology is tele-education. This thesis is an invitation to leave the comfort zone and question the usage of methodologies that have been highly proven with previous technologies, but which may as well be insufficient to address the challenges of new communications.

KEYWORDS

Quality of Experience, Video Quality, Subjective Assessment, Presence, Social Presence, Spatial Presence, Empathy, Attitude, Attention, Virtual Reality, Immersive Communications, 360° video, 360-degree video

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1 CONTEXT AND MOTIVATION

Virtual Reality (VR) is in constant development and improvement in terms of hardware and software in order to offer relevant experiences to users [1]. Initially, the greatest interest was generated in the entertainment sector. However, the latest trend in the commercialization of VR systems is oriented towards other sectors such as education, health, industry or commerce. One of the advances in VR has been the introduction of real visual information in the virtual environment, offering a more natural interaction with the

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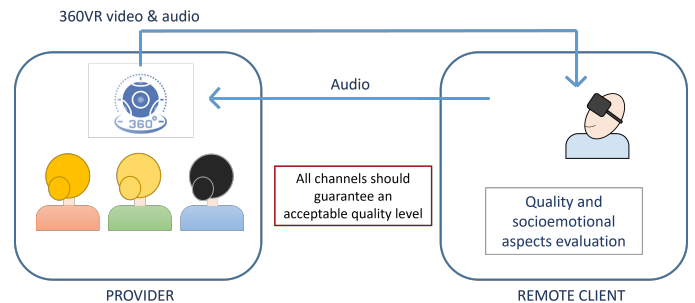


Figure 1: Immersive communication environment

scene and a greater acceptance of technology. Another advance is achieved with the representation of the scene through a video that covers the entire environment, called 360-degree video or omnidirectional video. This type of videos is acquired by cameras with omnidirectional lenses that cover the 360-degrees of the scene and is generally viewed by users through a Head Mounted Display (HMD). Users only visualize a small area of the 360-degree scene, called viewport, and the viewport changes with the variations of the viewing direction of the users, determined by the movements of the head [11].

In this thesis, we go one step further and consider that the displayed 360-degree video is being acquired synchronously elsewhere, allowing a real-time 360-degree video communication for teleconferencing, remote collaboration or teleoperation purposes. Figure 1 presents our scenario. We envision that this kind of communication will become mainstream within the next couple of decades. Our target is to research the technology that could make this possible, and design a proper evaluation methodology that scales for massive usage. Therefore, it is necessary to guarantee an acceptable Quality of Experience (QoE) to increase the use of immersive communications. The factors that influence the QoE can be divided into three categories: human (vision and hearing, simulator sickness, immersion, and expectations and expertise), system (content, network conditions, and encoding parameters) and context (physical, temporal, social, and task context) [17]. Based on this, cross-sectional research is necessary in the 360-degree video communications paradigm.

From the system perspective, 360-degree videos imply higher bitrates than 2D videos due to the resolution and framerate requirements. The pixels are distributed in a 360-degree scene, which requires at least 4K resolution to offer an acceptable quality. Also, the same framerate as the HMD refresh rate is strongly recommended, generally higher than the framerate used in 2D content. Despite the great effort of the research community to offer the

best quality to the user who visualizes the 360-degree video with the HMD, it is still a challenge to solve if we compare it with the quality that we perceive on a screen. On the contrary, features such as the ability to vary the area of the scene to be viewed, help the user to have a more realistic experience. Then, the advantages of this technology lie in helping the remote client to get closer to the people of the meeting and make this communication more effective thanks to the perception of non-verbal signals in a realistic way than current communication platforms [4].

The main problem we found is that the experiments carried out are based on the knowledge acquired with traditional contents [18–20] and recently adapted to immersive video [5]. These methodologies are designed to evaluate the perceived effect of the degradations caused along the signal processing chain (acquisition, compression, transmission, display...).

They consider short-duration videos (10s-20s) encoded with different qualities and displayed in random order to be evaluated, which implies that there is no narrative or context. Additionally, these methodologies do not consider the place where the 360-degree camera is placed or the acquisition perspective of the video (actor or first person or observer or third person). Hence, the main characteristics that influence on the human perspective, mainly determined by psychological effects induced in the users [22] that we called socioemotional aspects, are not fully considered. For example, presence, "the sense of being there" [21], empathy, "the ability to view the world from another person's perspective combined with an emotional reaction to that perspective, including feelings of concern for others" [2] or attitude, "an individual's evaluative judgment of the target behavior on some dimension" [6]. In this sense, a great variety of works can be found in the literature that demonstrate the effectiveness of this technology applications for achieving a higher enjoyment and involvement of users than traditional technologies [3, 10] and even the benefits on memory and attention assessments [23, 24]. Hence, we easily find questionnaires, environments, and applications that have been developed adjusted to the needs of each use case and experiment, usually in controlled environments and without considering the technical aspects.

Here, we found two main problems: reproducibility and ecological validity of the research. First, this kind of experiments is carried out with non open-source applications and with reduced samples of participants with specific characteristics. Second, the controlled environment limits the generalization of the findings in the experiments to the real world outside the laboratory [8]. Following the Unified Theory of Acceptance and Usage of Technology (UTAUT) [6], the acceptance of a technology by population is influenced by the performance expectancy and the effort expectancy. Based on the gaps that we find in the literature regarding the assessment of this technology, consistent methodologies, more realistic scenarios and a greater diversity of samples are necessary to ensure that this technology is accepted in society.

For this purpose, our starting point is a tele-education scenario. Before COVID-19, this technology was already a solution for those cases in which students must remain isolated or for those in rural areas who can not access school for weeks. However, the pandemic has caused millions of children and adolescents to suffer confinement situations that have kept them away from their daily life at school. As a consequence, some students have lost their rhythm

in academic terms and others have developed other psychological problems, motivated mainly by social distancing with their colleagues and teachers [7].

My hypothesis is that immersive communications can be a solution for the current situation, but a) methodologies that consider technical and socioemotional aspects and the influences between them, b) methodologies that consider different conditions (e.g. acquisition perspective), and c) the validation of these methodologies in real scenarios are necessary to increase the reliability of the assessments.

2 SPECIFIC RESEARCH OBJECTIVES, GOALS OR QUESTIONS

The main objective of this thesis is to provide a methodology to evaluate real-time 360-degree video communications in real scenarios. For that, we first present the research questions and then choose the tools, stimuli, and use case to answer them. We are interested in exploring the main questionnaires and biosensors used to evaluate QoE. With this, we want to propose a methodology, applicable to a communications environment and used with different conditions, that considers technical and socioemotional features. Finally, we propose the validation of the proposed methodology and the technology in real communications (non-controlled environments). Here, we present the stages that we tackle in this thesis:

- Stage 1: **Analysis of the methodologies and tools to evaluate video quality and socioemotional aspects**
- Stage 2: **Analysis of conditions that influence the socioemotional aspects**
- Stage 3: **Experiments with bidirectional communications in a real use case, tele-education**

The stages 1 and 2 of this thesis look for the validation of different parts of the methodology and tools. Finally, we propose the integration of all of them to carry out the stage 3 which covers experiments in real use cases with bidirectional communications and non-controlled environments to prove the benefits of this type of communications. So, once we have proposed a methodology and we have tested it in laboratory assessments, we take it to a real environment. As a starting point, we choose tele-education use case, in which the remote student attends the lesson using an HMD device to visualize the 360-degree scene of his/her classroom.

2.1 Challenges

If we want to ensure that this type of communication is used instead of computer or mobile video conferencing, we need methodologies that leave no one out and work in real scenarios. So, the main challenges presented in this thesis are:

- How can we jointly evaluate quality and socioemotional aspects?
- What conditions such as video quality or acquisition perspective affect in terms of socioemotional aspects in an immersive communication?
- Can biosensors help to evaluate QoE in an immersive communication?
- How different scales designed to evaluate the performance of an immersive communication can be compared?

- Is it possible to design a methodology that can be applied to real-time 360-degree video communications? How could it be validated?

3 RESEARCH PROGRESS TO DATE

Stages 1 and 2 have been mostly completed apart from the ongoing research stay focused on the exploration of the use of biosensors in the evaluation of socioemotional aspects and QoE. The final step is Stage 3, experiments with bidirectional communication in tele-education scenario.

4 RESULTS AND CONTRIBUTIONS TO DATE

This thesis started with the first stage, totally related to our background and experience. We performed an exhaustive analysis of the traditional objective metrics and the adaptations of them for omnidirectional contents. Specifically, one relevant work was the validation of Video Multimethod Assessment Fusion (VMAF) [12], a full reference objective metric developed by Netflix initially designed to work with traditional 2D contents, on 360VR contents [13]. In this work, we assumed that VMAF scores decrease monotonically with the Quantization Parameter (QP) due to the effect of this encoding parameter in the quality of the sequence. Then, the process was divided into two steps. Firstly, we directly applied the objective metric to obtain the VMAF score of a wide variety of 360-degree videos encoded with constant QP in the whole range of possible values (1-51), obtaining the relation between QP and VMAF. Secondly, this relation of QP-VMAF was validated with a subjective assessment. For that, we selected a finite number of VMAF scores, operating points sufficiently spaced, per content and this subset of encoded 360-degree videos was evaluated by participants. We normalized the participant evaluations calculating the Differential Mean Opinion Scores (DMOS). The high correlation between the two curves allows us to conclude that VMAF works correctly with 360-degree videos homogeneously encoded, without modifying the implementation of VMAF. This result is an important contribution since offering an acceptable video quality is necessary to offer users a good QoE and improve socioemotional aspects such as immersion. Likewise, it is a very useful tool for the research community focused on the development of techniques and encoding schemes to save bitrate by offering the best possible quality.

As a first step to jointly assess video quality and socioemotional aspects, we carried out a pilot study where video quality and sense of presence were evaluated in two of the most popular HMDs (Samsung S8 + Samsung GearVR which includes a touchpad on its right side and Lenovo Mirage Solo with a handheld controller) [15]. Video quality was evaluated using the ACR-HR (Absolute Category Rating with Hidden Reference) methodology with a five-level rating scale, as recommended in ITU-T P.910 [19]. Presence was assessed with two of the highly tested questionnaires: the Temple Presence Inventory (TPI) [9] and the Presence Questionnaire (PQ) [25]. As a result, we provided a repository that contains: a dataset of video sources with the associated objective metrics and other specifications, head tracking data and rates (video quality and presence), and statistical analysis notebook. The main conclusions were a relevant input for the design of posterior experiments. The visualization of repeated short-duration clips decreases the sense of presence motivated by

what we called fatigue effect. It was a relevant motivation to explore in depth methodologies that cover the evaluation of video quality and socioemotional aspects and the interaction between them. Additionally, thanks to the informal inputs about the influence of the 360-degree video on the sense of presence, we decided to explore higher-level aspects such as acquisition perspective, camera location, and interactive elements in the following experiments. Also, we concluded that the controller is a better evaluation method in terms of comfort for participants. As member of Video Quality Experts Group (VQEG), the conclusions of this pilot study were a contribution to create ITU-T Recommendation P.919 for the quality evaluation of VR, a new standard for subjective test methodologies for 360-degree video on HMDs [5].

A more complex methodology to assess video quality, spatial and social presence, empathy, attitude, and attention in 360-degree videos for immersive communications was proposed [16]. We validated it in an experiment where the immersive communication was simulated. Then, participants visualized three contents, designed and acquired for the experiment, considering conversations of different genre (everyday conversation, educational, and discussion) and actor and observer acquisition perspectives. Videos were encoded with quality fluctuations, simulating a streaming session with unstable network conditions. The conversations were placed in the context of international experiences to have a diverse sample of participants with experiences working or studying in a foreign country. We consider three experimental conditions: (A) visualizing and rating the perceptual quality of contents in an HMD, (B) visualizing the contents in an HMD, and (C) visualizing the contents in an HMD where participants can see their hands and take notes. Video quality was evaluated with Single-Stimulus Discrete Quality Evaluation (SSDQE) methodology. Spatial and social presence were evaluated with questionnaires adapted from the literature. Initial empathy was assessed with Interpersonal Reactivity Index (IRI) and a questionnaire was designed to evaluate the attitude after the visualization of each video. Attention was evaluated with three questions that had pass/fail answers. As a contribution, we highlight the proposal of a methodology to assess socioemotional features with the additional task of evaluating the video quality. Also, we highlight the fact that the evaluation of the video quality and the socioemotional aspects influence the results, increasing the interest in validating the methodology in more realistic scenarios.

In relation to the context of tele-education, chosen as starting point of the stage 3, we created EVENT-CLASS, a dataset of events in the classroom [14]. We acquired sequences of 360-degree video with audio considering a great variety of environments, lighting conditions, acquisition perspectives, and cameras (qualities) in the context of education. Also, people and students raising their hands during the lesson have been annotated in the sequences. The main goal of the creation of EVENT-CLASS was to provide a rich 360-degree database to be used in quality assessment tests and in the implementation of machine learning techniques. In this sense, Kachach et al. [7] presents the overview of the system where the methodology will be applied and validated as last step of the thesis.

5 EXPECTED NEXT STEPS

The main next steps are focused on the review of the questionnaires already designed, and the exploration of the use of biosensors to obtain a solid methodology. Once the methodology is well defined and tested in controlled environments, the work will focus on beginning to validate it in non-controlled assessments in schools, in which this technology and communications is a solution for tele-education. The plan is to carry out the assessments at the beginning of the 2022/2023 academic year in Spain, September 2022.

6 DISSERTATION STATUS AND LONG TERM GOALS

I am a second year PhD student in the Image Processing Group-*Grupo de Tratamiento de Imágenes (GTI) at the Universidad Politécnica de Madrid (UPM)*. I have already completed a literature review, some pilot studies, and carried out formal experiments in which the technical side of the methodology has already been validated. Currently, I am doing a research stay in the *Humane Technology Laboratory at the Università Cattolica del Sacro Cuore*. This research stay is a transversal knowledge for my thesis since it complements the methodology and is necessary to design and validate the non-technical part of my thesis. Phase 1, and 2 have been mostly completed, apart from the ongoing exploration of the use of biosensors in the QoE experiments that is planned to be completed by May 2022. My final step is the stage 3 in which the methodology is applied in a real tele-education scenario where bidirectional communications are provided.

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